WEEK 6 - Product Review And Rating Management System

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Scenario:

The Product Review and Rating Management System is a database application used to collect and manage customer feedback about purchased products. It allows customers to share their opinions and rate products based on quality, design, durability, and overall satisfaction.

This system connects customer information with product details and helps identify popular products and areas that require improvement. It also helps businesses understand customer preferences and improve product quality.

Review and Rating Table Creation A Review table is used to store customer feedback and product ratings.

It contains details such as Review ID, Customer ID, Product ID, Rating, Review Comment, and Review Date. A rating constraint ensures that customers can provide ratings only between 1 and 5, where 1 represents very poor satisfaction and 5 represents excellent satisfaction.

```
SQL> connect system /
Enter password:
Connected.
SQL> connect
Enter user-name: c##sownd
Enter password:
Connected.
SQL> CREATE TABLE Review (
  2      Review_ID VARCHAR2(5) PRIMARY KEY,
  3      Customer_ID VARCHAR2(5),
  4      Product_ID VARCHAR2(7),
  5      Rating NUMBER(1) CHECK (Rating BETWEEN 1 AND 5),
  6      Review_Comment VARCHAR2(200),
  7      Review_Date DATE,
  8      FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID),
  9      FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
 10  );

Table created.

SQL> DESC Review;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------
----
 REVIEW_ID                                 NOT NULL VARCHAR2(5)
 CUSTOMER_ID                                        VARCHAR2(5)
 PRODUCT_ID                                         VARCHAR2(7)
 RATING                                             NUMBER(1)
 REVIEW_COMMENT                                     VARCHAR2(200)
 REVIEW_DATE                                        DATE
```

Inserting values

```
SQL> INSERT INTO Review
  2  VALUES ('R101', 'C001', 'P001', 5, 'Excellent performance', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R102', 'C002', 'P002', 4, 'Good quality product', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R103', 'C001', 'P002', 4, 'Good product', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R104', 'C002', 'P001', 3, 'Average product', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R105', 'C001', 'P001', 5, 'Excellent product', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R106', 'C002', 'P002', 5, 'Very good quality', SYSDATE);

1 row created.
```

```
  2  VALUES ('R107', 'C001', 'P002', 3, 'Product is satisfactory', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R108', 'C002', 'P001', 4, 'Worth the price', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R109', 'C001', 'P001', 5, 'Highly recommended', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R110', 'C002', 'P002', 2, 'Quality can be improved', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R111', 'C001', 'P002', 4, 'Nice product', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R112', 'C002', 'P001', 5, 'Amazing product', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R113', 'C001', 'P001', 3, 'Good but expensive', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO Review
  2  VALUES ('R114', 'C002', 'P002', 4, 'Satisfied with product', SYSDATE);

1 row created.
```

Activate Windows
Go to Settings to activate Wi

Updating reviews:

Modify review comments when necessary.

Deleting reviews:

Remove inappropriate, invalid, or unwanted reviews

```
SQL> UPDATE Review
  2  SET Review_Comment = 'Excellent product quality'
  3  WHERE Review_ID = 'R101';

1 row updated.

SQL>
```

```
SQL> DELETE FROM Review
  2  WHERE Review_ID = 'R106';

1 row deleted.

SQL>
```

Retrieving All Product Review Details Retrieving review details helps businesses understand customer opinions about their products.

The database can display all reviews associated with a particular product and show customer names along with their comments and rating.

```
SQL> SELECT * FROM Review;

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
R101  C001  P001             5
Excellent performance
20-OCT-26

R102  C002  P002             4
Good quality product
20-OCT-26

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------

R103  C001  P002             4
Good product
20-OCT-26

R104  C002  P001             3
Average product

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
20-OCT-26

R105  C001  P001             5
Excellent product
20-OCT-26

R106  C002  P002             5
```

```
REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------
REVIEW_DA
---------
Very good quality
20-OCT-26

R107  C001  P002             3
Product is satisfactory
20-OCT-26


REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------
REVIEW_DA
---------
R108  C002  P001             4
Worth the price
20-OCT-26

R109  C001  P001             5
Highly recommended
20-OCT-26

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------
REVIEW_DA
---------

R110  C002  P002             2
Quality can be improved
20-OCT-26

R111  C001  P002             4
Nice product

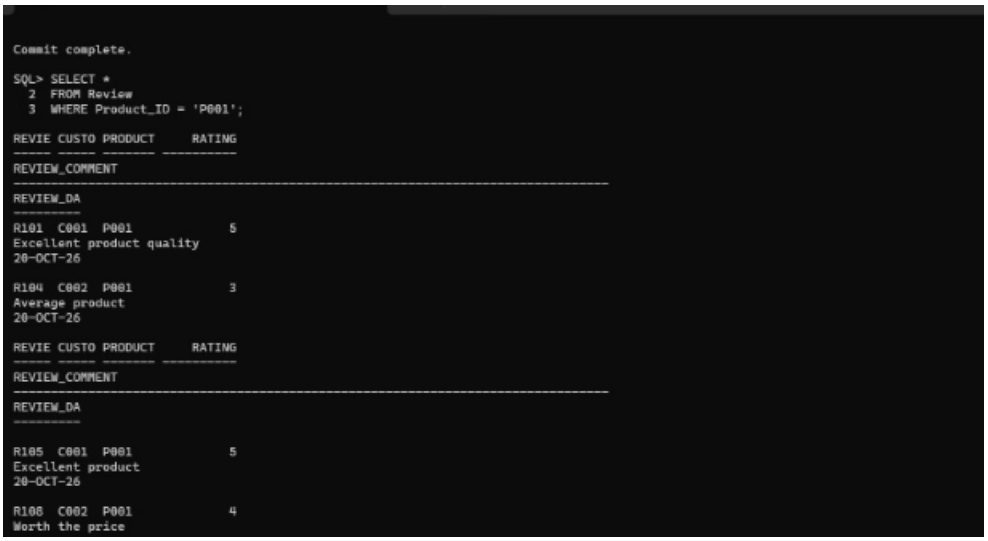
REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------
REVIEW_DA
---------
20-OCT-26
```

```
REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------
REVIEW_DA
---------
R115  C001  P001             5
Best product
20-OCT-26


15 rows selected.
```

❖ Retrieve Product Review Details

1. Display all reviews for a produc



```
Commit complete.

SQL> SELECT *
  2  FROM Review
  3  WHERE Product_ID = 'P001';

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
R101  C001  P001            5
Excellent product quality
20-OCT-26

R104  C002  P001            3
Average product
20-OCT-26

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------

R105  C001  P001            5
Excellent product
20-OCT-26

R108  C002  P001            4
Worth the price
```

2. Display customer name with their reviews

```
Connected.
SQL> SELECT C.Customer_Name,
  2         R.Review_Comment,
  3         R.Rating
  4  FROM Customer C
  5  JOIN Review R
  6  ON C.Customer_ID = R.Customer_ID;

CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
    RATING
----------
Rahul
Excellent product quality
         5

Ananya
Good quality product
         5

CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
    RATING
----------

Rahul
Good product
         4

Ananya
Average product

CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
    RATING
----------
         3

Rahul
Excellent product
         5

Rahul

CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
    RATING
----------
Product is satisfactory
         3

Ananya
Worth the price
         4


CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
```

```
CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
    RATING
----------
Rahul
Highly recommended
         5

Ananya
Quality can be improved
         2

CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
    RATING
----------

Rahul
Nice product
         4

Ananya
Amazing product

CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
    RATING
----------
         5

Rahul
Good but expensive
         3

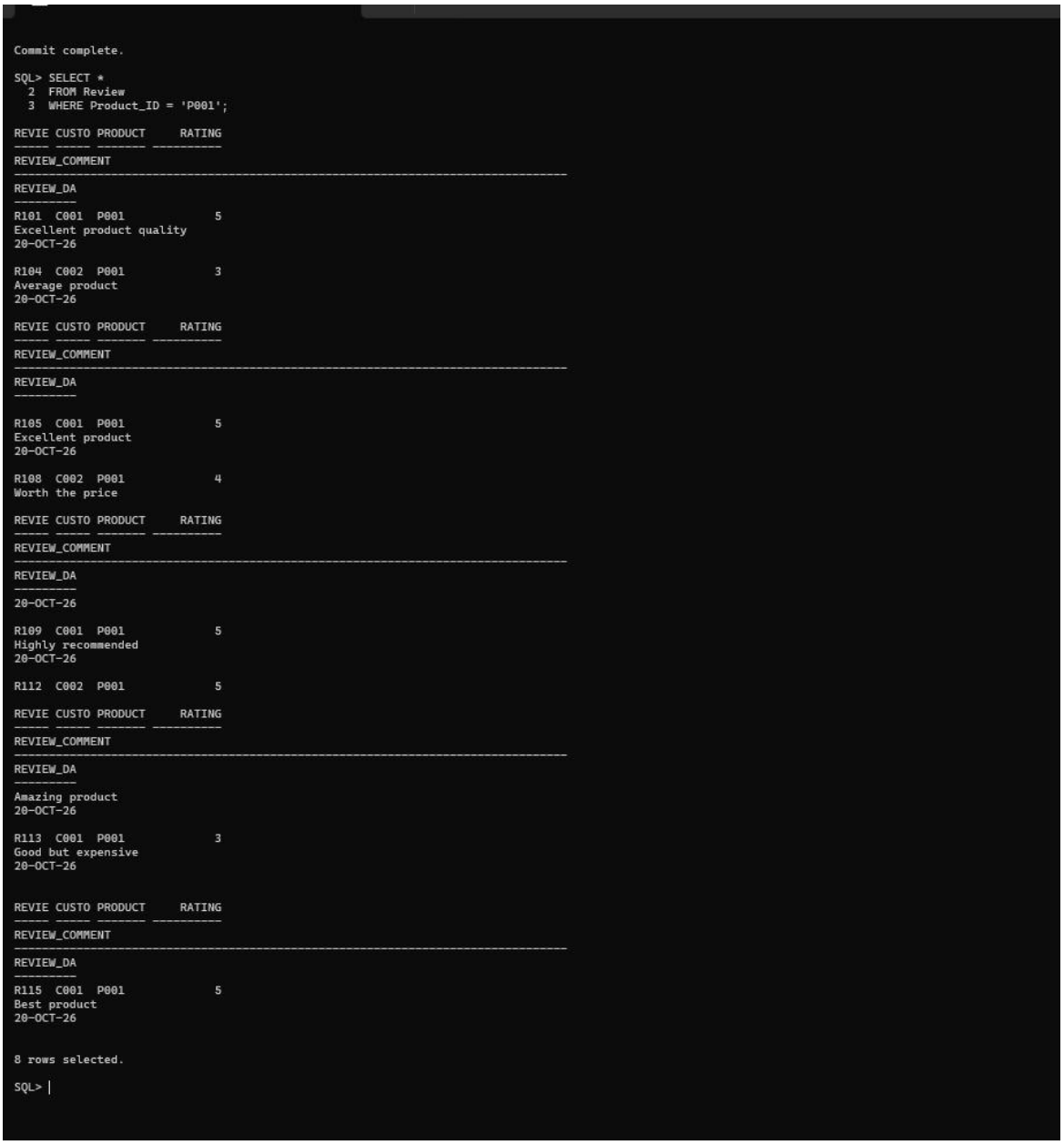
Ananya

CUSTOMER_NAME
--------------------------------------------------
REVIEW_COMMENT
--------------------------------------------------------------------------------
    RATING
----------
Satisfied with product
         4

Rahul
Best product
         5


14 rows selected.
```

3. Find products having maximum reviews



```
Commit complete.

SQL> SELECT *
  2  FROM Review
  3  WHERE Product_ID = 'P001';

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
R101  C001  P001             5
Excellent product quality
20-OCT-26

R104  C002  P001             3
Average product
20-OCT-26

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------

R105  C001  P001             5
Excellent product
20-OCT-26

R108  C002  P001             4
Worth the price

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
20-OCT-26

R109  C001  P001             5
Highly recommended
20-OCT-26

R112  C002  P001             5

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Amazing product
20-OCT-26

R113  C001  P001             3
Good but expensive
20-OCT-26


REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
R115  C001  P001             5
Best product
20-OCT-26


8 rows selected.

SQL>
```

4. Retrieve reviews with ratings above 4

```
P001                8

SQL> SELECT *
  2  FROM Review
  3  ORDER BY Review_Date DESC;

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
R115  C001  P001             5
Best product
20-OCT-26

R102  C002  P002             5
Good quality product
20-OCT-26

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------

R103  C001  P002             4
Good product
20-OCT-26

R104  C002  P001             3
Average product

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
20-OCT-26

R105  C001  P001             5
Excellent product
20-OCT-26

R107  C001  P002             3

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Product is satisfactory
20-OCT-26

R109  C001  P001             5
Highly recommended
20-OCT-26
```

```
R113  C001  P001            3
Good but expensive

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
20-OCT-26

R114  C002  P002            4
Satisfied with product
20-OCT-26

R101  C001  P001            5

REVIE CUSTO PRODUCT     RATING
----- ----- ------- ----------
REVIEW_COMMENT
--------------------------------------------------------------------------------
REVIEW_DA
---------
Excellent product quality
20-OCT-26

R108  C002  P001            4
Worth the price
20-OCT-26


14 rows selected.

SQL> |
```

❖ Calculate Average Product Ratings

1. Average rating for each product

```
SQL> SELECT Product_ID,
  2         ROUND(AVG(Rating), 2) AS Average_Rating
  3  FROM Review
  4  GROUP BY Product_ID;

PRODUCT AVERAGE_RATING
------- --------------
P001              4.38
P002              3.67
```

2. Count number of reviews per product

```
SQL> SELECT Product_ID,
  2         COUNT(*) AS Number_Of_Reviews
  3  FROM Review
  4  GROUP BY Product_ID;

PRODUCT NUMBER_OF_REVIEWS
------- -----------------
P001                    8
P002                    6

SQL>
```

3. Find highest-rated products

```
SQL> SELECT Product_ID,
  2         ROUND(AVG(Rating), 2) AS Average_Rating
  3  FROM Review
  4  GROUP BY Product_ID;

PRODUCT AVERAGE_RATING
------- --------------
P001              4.38
P002              3.67
```

4. Products with average rating above 4

```
SQL> SELECT Product_ID,
  2         ROUND(AVG(Rating), 2) AS Average_Rating
  3  FROM Review
  4  GROUP BY Product_ID
  5  HAVING AVG(Rating) > 4;

PRODUCT AVERAGE_RATING
------- --------------
P001              4.38

SQL>
```

Reports Report 1:

Product Rating Analysis The Product Rating Analysis report provides an overview of product performance.

It includes product names, the number of reviews received, and the average rating of each product.

This report helps compare products and identify those that receive positive customer feedback.

```
SQL> SELECT P.Product_Name,
  2         COUNT(R.Review_ID) AS Number_Of_Reviews,
  3         ROUND(AVG(R.Rating), 2) AS Average_Rating
  4  FROM Product P
  5  JOIN Review R
  6  ON P.Product_ID = R.Product_ID
  7  GROUP BY P.Product_Name;

PRODUCT_NAME
--------------------------------------------------------------------------------
NUMBER_OF_REVIEWS AVERAGE_RATING
----------------- --------------
Samsung Case
                6           3.67

iPhone Case
                8           4.38
```

• Highly rated product

```
SQL> SELECT Product_ID, AVG(Rating) AS Average_Rating
  2  FROM Review
  3  GROUP BY Product_ID
  4  HAVING AVG(Rating) > 4;

PRODUCT AVERAGE_RATING
------- --------------
P001             4.375

SQL>
```

• Products requiring improvement:

```
SQL> SELECT Product_ID, AVG(Rating) AS Average_Rating
  2  FROM Review
  3  GROUP BY Product_ID
  4  HAVING AVG(Rating) < 3;

no rows selected

SQL>
```

Report 3: Rating Distribution

The Rating Distribution report shows how ratings are distributed across products. It includes the number of 5-star ratings, the number of 4-star ratings, lower-rated products, and the overall distribution of ratings from 1 to 5.

This report provides a clear picture of customer satisfaction and helps identify products that need attention.

• Number of 5-star ratings

```
SQL> SELECT COUNT(*) AS Five_Star_Ratings
  2  FROM Review
  3  WHERE Rating = 5;

FIVE_STAR_RATINGS
-----------------
                6
```

• Number of 4-star ratings

```
SQL> SELECT COUNT(*) AS Four_Star_Ratings
  2  FROM Review
  3  WHERE Rating = 4;

FOUR_STAR_RATINGS
-----------------
                4
```

• Number of low-rated reviews

```
SQL> SELECT COUNT(*) AS Low_Rated_Reviews
  2  FROM Review
  3  WHERE Rating <= 2;

LOW_RATED_REVIEWS
-----------------
                1
```

• Complete rating distribution

```
SQL> SELECT Rating, COUNT(*) AS Number_Of_Reviews
  2  FROM Review
  3  GROUP BY Rating
  4  ORDER BY Rating DESC;

    RATING NUMBER_OF_REVIEWS
---------- -----------------
         5                 6
         4                 4
         3                 3
         2                 1
```